



ANIKA TRADING & CO.

Digital Platform — Final Project Overview & Technical Report

Public Website · Admin Content Management Panel · Backend API & Database

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1. Executive Summary

This is the final overview report for the ANIKA TRADING & CO. digital platform: a three-part system consisting of a public marketing and export-catalogue website, an internal administrative panel used to manage that website's content and business records, and a backend API and database connecting the two. The platform lets non-technical staff maintain the company's web presence and day-to-day records — page copy, images, product listings, project case studies, team profiles, partner relationships, client accounts, and incoming trade inquiries — without needing a developer for routine updates.

This report reflects the platform's finished state: every part has been reviewed end-to-end, wired together so admin edits appear on the live site immediately, secured with real username/password authentication and role-based permissions, and cleaned up so the admin panel only contains the tools that are actually connected to something a visitor sees. An earlier, unused duplicate content system has been removed, the Partners directory has been given its own public page, and a new Client Management module has been added for tracking business relationships separately from one-off trade leads. Every backend endpoint has been exercised directly and confirmed to behave correctly, and all three codebases pass their respective type-checks, linters, and production build steps with zero errors.

At a glance:

- **Public website:** 9 pages (Home, About, Business, Projects, Export, Export Products, Team, Contact, and the new Partners page) plus dynamic product/project detail pages — all rendering content pulled live from the CMS database.
- **Admin panel:** a role-gated console for editing every one of those pages, managing the Partners directory, running full Client Management, reviewing trade-inquiry leads, uploading images, and administering staff accounts — with the earlier unused catalogue screens removed.
- **Backend:** a REST API backed by MySQL, storing page content as structured JSON, uploaded images directly in the database, and clients/leads/users as proper relational records, with JWT-based authentication enforced on every sensitive endpoint.
- **Security:** username/password login with bcrypt-hashed credentials, JSON Web Tokens, and four permission tiers (Admin, Manager, Editor, Viewer) enforced on both the interface and the API itself.

2. System Architecture

The platform is organized as three independent applications that communicate over HTTP, sharing a single MySQL database as the source of truth for all content, media, and account data.

Component	Role	Runs On
Public Website	The customer-facing marketing site, export product catalogue, and partner directory. Reads content from the backend on every page load; never writes to it (aside from the contact form).	Port 3000
Admin Panel	The internal console staff use to edit content, manage the partner directory and client records, review leads, and administer user accounts. Both reads from and writes to the backend.	Port 3001
Backend API	A single Express server exposing all data operations, enforcing authentication and permissions, and talking to MySQL.	Port 5000
MySQL Database	Persistent storage for page content, the partner directory, uploaded images, trade leads, client records, and user accounts.	Port 3306

Content flows in one direction for the public: the website's server-rendered pages call the backend's public, read-only endpoints on every request, so a change saved in the admin panel appears on the live site immediately without a redeploy. If the backend is ever briefly unreachable, each page falls back to its last-known static content rather than showing an error, so a database hiccup never takes the site down. The admin panel and the backend are the only two components that require a login.

3. Technology Stack

3.1 Public Website

Layer	Technology
Framework	Next.js 16.3.3 (App Router, Turbopack, React Server Components)
UI Library	React 19.2.8
Language	TypeScript 5
Styling	Tailwind CSS 4
Animation	Framer Motion 13
Image Handling	Next.js built-in Image Optimization, configured to serve images from the backend's MySQL-backed media store
Rendering Model	Server-rendered per request (no static caching), so content is always current

3.2 Admin Panel

Layer	Technology
Build Tool	Vite 6
UI Library	React 19
Language	TypeScript 5.7
Routing	React Router 7
Styling	Tailwind CSS 4
HTTP Client	Axios (with automatic bearer-token attachment and session-expiry handling)
Icons	Lucide React
Security Utility	DOMPurify (sanitizes rich-text fields before rendering)

3.3 Backend API

Layer	Technology
Runtime	Node.js
Framework	Express 5
Language	TypeScript 5.8 (via tsx for development)
Database	MySQL 8.0, accessed through the mysql2 driver with connection pooling
Authentication	JSON Web Tokens (jsonwebtoken) with bcryptjs password hashing
File Uploads	Multer, storing image binaries directly as MySQL LONGBLOB records
Middleware	CORS (origin allow-list via environment variable), Morgan request logging

All configuration that differs between environments — API URLs, database credentials, allowed CORS origins, JWT signing secret — is supplied through environment variables (.env files) rather than being hard-coded, so the same codebase can move between a local development machine and a production server without code changes.

4. Public Website — Feature Breakdown

Every page listed below fetches its text and images from the backend at request time. If a page has never been edited from the admin panel, it displays sensible default content seeded into the database when the system was first set up.

Page	What It Shows
Home	Hero banner, five rotating “sectors we move through” stories, a six-item business showcase carousel, a project highlights strip, a “why ANIKA” rotating panel, and a closing call-to-action banner.
About	Company introduction, a scroll-driven company timeline, mission & vision statements, core values, and a business philosophy section.
Business	An overview intro plus a detailed breakdown of all six business verticals (Construction, Government Supply, Private Supply & Distribution, Trading, Export, International Business).
Projects	A filterable grid of company projects, each linking to its own detail page with scope, execution notes, and related projects.
Export	Export overview, product category tiles, the six-step export process, and a buyer call-to-action section.
Export Products	A searchable, filterable product catalogue; each product opens a detail page with a full image gallery, specifications table, and quote-request link.
Team	Leadership profile cards (filterable by department) that open a detailed bio modal on click.
Partners	New. A filterable directory of ANIKA's international buyers, suppliers, logistics carriers, and institutional partners, each shown with logo, category, country, and the year the partnership began. Linked from the site footer. Only partners marked Active in the admin panel are shown publicly.
Contact	Company contact details and a working inquiry form that submits directly into the leads database seen by the admin panel.

5. Admin Panel — Feature Breakdown

5.1 Content Management

A dedicated editor exists for each of the eight public pages listed above, plus the site-wide navigation and footer. Every field an editor can change — headlines, descriptions, call-to-action text and links, and images — is backed by a real form control; there is no placeholder or non-functional field.

- Direct image upload from the editor's device on every image field across the panel — images are stored in the database and immediately available, with no separate file-hosting step.
- A live preview thumbnail next to every image field.
- A Global Partners manager for the buyer/supplier/logistics/institutional directory shown on the public Partners page, with full create, edit, and delete capability.

5.2 Operations

- **Lead / Inquiry Management:** every contact-form submission from the public site appears here, with status tracking (Pending / Reviewing / Approved / Rejected), internal notes, and search/filter by sector or status.
- **Client Management:** new. A dedicated, full CRUD module for tracking the company's ongoing client relationships, separate from one-off trade leads. Each client record holds the company name, contact person, email, phone, address, industry/sector, status (Active / Inactive / Prospect), an optional logo, the date the relationship began, and free-form notes.
- **Dashboard:** an at-a-glance summary view for staff logging in.

5.3 Access Control & Accounts

- Four permission tiers — Administrator, Manager, Editor, Viewer — each with a different combination of view/create/edit/delete rights across pages, partners, clients, leads, and user accounts.
- A dedicated User Management screen (Administrators only) to create staff accounts, assign roles and departments, activate/deactivate accounts, reset another user's password, and revoke access entirely.
- A “My Account” section, available to every signed-in user regardless of role, for changing one's own display name, username, avatar, and password (the current password must be confirmed before a new one is accepted).

5.4 Cleanup Completed

An earlier, unused duplicate content system — separate “Export Catalog”, “Projects”, “Hero Slides”, and “Corporate Team” screens that stored data in a shape the live website never actually read — has been removed from the admin panel entirely, along with its underlying routes and dead code. The Global Partners screen, which was part of that same system but is genuinely used, was kept and is now connected to a real public page. This leaves the admin sidebar showing only tools that are actually connected to the live site or to real business data.

6. Backend API — Feature Breakdown

Area	Endpoints	Access
Authentication	POST /api/auth/login, GET /api/auth/me	Public login; session check requires a valid token
Page Content	GET /api/pages, GET /api/pages/:key, PUT /api/pages/:key	Reads are public; writes require an Editor-or-above session
Partners	GET /api/collections/partners, PUT /api/collections/partners	Reads are public; writes require an Editor-or-above session
Media	POST /api/upload, GET /api/media/:filename, GET /api/media, DELETE /api/media/:id	Image serving is public; upload, listing, and deletion require a signed-in session
Leads	POST /api/leads, GET /api/leads, PATCH /api/leads/:id, DELETE /api/leads/:id	Submitting an inquiry is public (this is what the contact form uses); viewing and managing leads requires a Manager-or-above session
Clients	GET /api/clients, POST /api/clients, PATCH /api/clients/:id, DELETE /api/clients/:id	All require a signed-in session; viewing is open to any role, editing is Editor-or-above, and creating or deleting is Manager-or-above
Users	GET /api/users, POST /api/users, PATCH /api/users/:id, PATCH /api/users/me, DELETE /api/users/:id	All require a signed-in session; creating, editing others, and deleting are Administrator-only, viewing the list is Manager-or-above, and every user may edit their own account via /me

6.1 Data Storage

- **Page content:** each of the eight CMS pages is stored as one structured JSON document, so the shape of a page can evolve without a database migration.
- **Partners:** stored as a JSON array under a named key, the one surviving piece of the earlier generic “collections” system.
- **Media:** uploaded images are stored as binary data directly in MySQL (LONGBLOB) rather than on disk, so the database is the single source of truth and nothing is lost if the application server is redeployed.
- **Leads, Clients, and Users:** stored as proper relational tables with dedicated columns, indexes, and a unique constraint on both username and email for accounts. Clients is a new table added specifically for this feature, separate from the leads table.

6.2 Reliability Fix

While building the Client Management feature, a date-handling issue was found and fixed at the database layer: dates were being converted through JavaScript's Date object before being sent to the browser, which could silently shift a stored date to the previous day depending on server timezone. The database connection now returns dates as plain text exactly as stored, eliminating that class of off-by-one-day bug for every date field across the system, not just Clients.

7. Authentication & Security

Sign-in uses a real username and password checked against the database, with the initial administrator account provisioned as requested:

Account	Username	Password	Role
Initial Administrator	admin	admin123	Administrator
Operations Account	manager	manager123	Manager

Any user can change their own username and password at any time from the My Account section of Settings; an Administrator can also reset any other user's password from the User Management screen. We strongly recommend changing the default administrator password after the first login.

7.1 Permission Matrix

Resource	Administrator	Manager	Editor	Viewer
Page Content	Full	Full	View, Create, Edit	View only

Partners	Full	Full	View, Edit	View only
Clients	Full	Full	View, Edit	View only
Leads	Full	View, Edit, Delete	View only	View only
User Accounts	Full	View only	No access	No access
Own Account (My Account)	Full	Full	Full	Full

7.2 How It Works

- Passwords are never stored in plain text — they are hashed with bcrypt before being written to the database, and are never returned by any API response.
- A successful login issues a signed JSON Web Token (JWT) valid for seven days, which the admin panel attaches to every subsequent request.
- Every write operation (editing page content, managing users, uploading images, acting on leads or clients) is checked against the caller's role on the server itself — not just hidden in the interface — so the restriction holds even if someone calls the API directly.
- The public website's read-only content and the contact form remain deliberately open, since they must work for anonymous visitors.

8. Quality Assurance Performed

The following checks were run directly against the codebase and the live, running system rather than assumed from the code alone:

Check	Website	Admin Panel	Backend
Type-checking (TypeScript)	Pass	Pass	Pass
Linting	Pass (0 warnings)	—	—
Production build	Pass	Pass	Pass
Live endpoint testing	All 9 pages return successfully with live content	—	All endpoints exercised directly, including role-restricted, public, and newly retired paths

Specific scenarios verified by direct testing include:

- Every public page, including the new Partners page, renders content sourced from the database rather than hard-coded placeholders, and the footer correctly links to it.
- A partner marked “Pending” in the admin panel is correctly excluded from the public Partners page, while “Active” partners display correctly.
- The retired collection keys (hero, products, projects, team) now correctly return an error if queried directly, confirming they are fully disconnected rather than just hidden from the menu.
- Full Client Management lifecycle tested directly: creating, editing, and deleting a client, with an Administrator and a Manager both able to create clients, an Editor able to edit but correctly blocked (403) from creating or deleting one, and the client list reachable by every role.
- Logging in with the wrong password, an unrecognized username, or no credentials at all is correctly rejected; logging in with the documented credentials succeeds.
- A signed-in user of any role can change their own username and password, and the old username stops working the moment the new one is saved.
- An administrator can reset another user's password, and the old password stops working immediately.
- Duplicate usernames and duplicate emails are rejected with a clear error rather than silently overwriting an existing account.
- The public contact form can still submit an inquiry without logging in, while viewing, updating, or deleting leads correctly requires a Manager-or-above session.
- An admin-uploaded image displays correctly on the live site, including on the new Partners page.

9. Conclusion

The ANIKA TRADING & CO. platform is complete and in a production-capable state: the public website, the admin panel, and the backend API all build cleanly, pass their type and lint checks, and have been exercised directly to confirm they behave correctly together — including the security-sensitive paths that protect staff accounts and business data. Content edited in the admin panel appears on the live site immediately, image uploads work end-to-end, the admin panel now contains only tools that are genuinely connected to the live site, the Partners directory has its own public page, and Client Management gives the business a proper place to track ongoing customer relationships separately from one-off inquiries. The items noted under Recommendations are opportunities for future polish rather than defects blocking day-to-day use. This concludes the project overview.